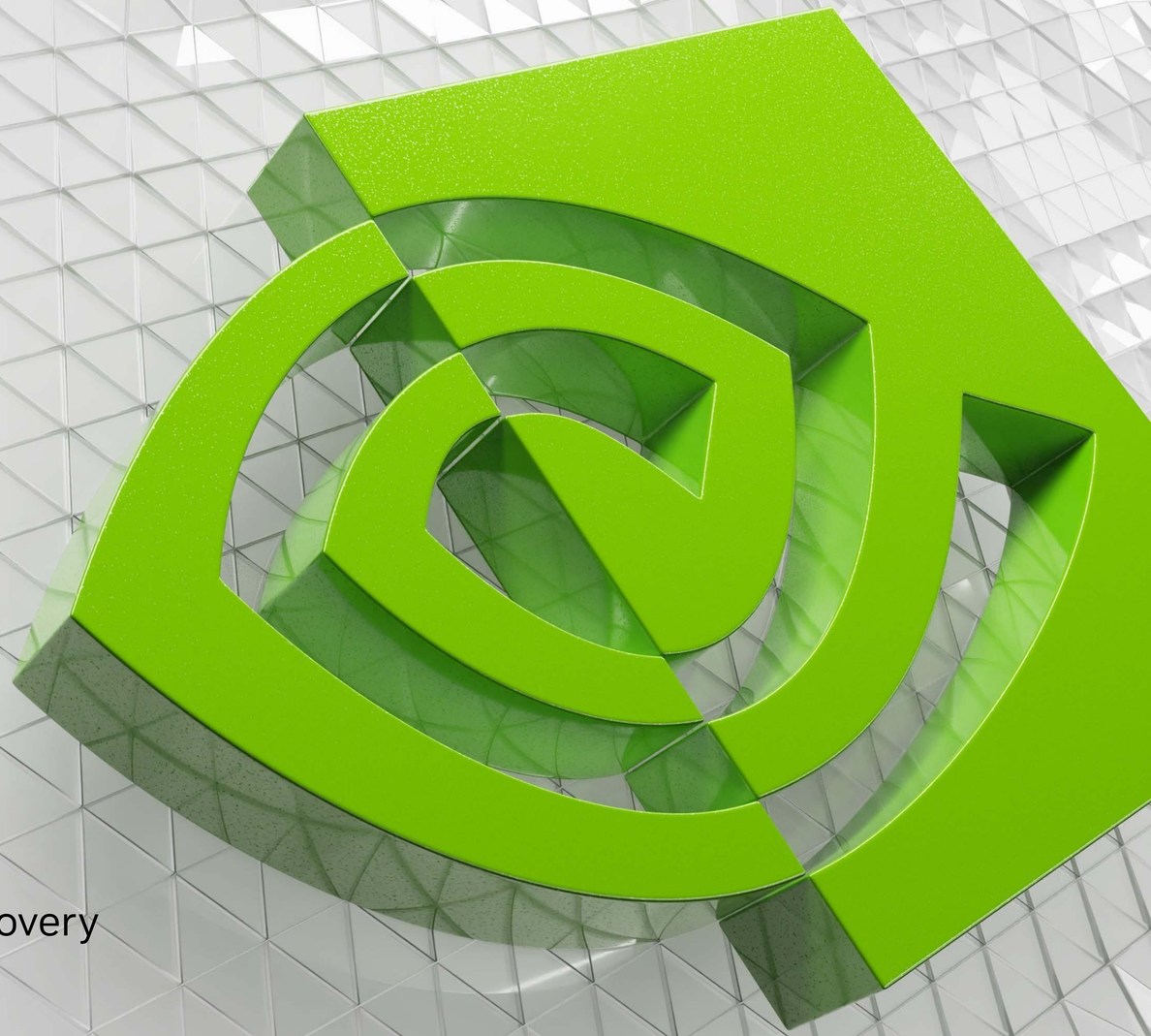
NVIDIA BioNeMo

The Development Platform for AI-driven Biology and Drug Discovery



Enabling Automation of Science

The Future of Self Driving Labs



Agentic AI

Hypothesis generation, protocol design, lab execution, data analysis

Design
Experiments, Candidates



Learn
Update Model and Parameters

Make
Molecules



Test
Biochemical Properties

NVIDIA BioNeMo Platform

Open Models, Data and Tools for Drug Discovery and Design

Datasets

Open Datasets

AlphaFold Database (30M Complexes)

Chemistry Reasoning (Litmus Bench)

Models

Target identification (CodonFM)

Protein Design (Proteina-Complexa)

Small Molecule Design (GenMol)

Property Prediction (KERMT)

Synthesizability (ReaSyn)

Libraries

Genomics secondary analysis (Parabricks)

cheminformatics (nvMolKit)

Systems pharmacology (nvQSP)

Single-cell analysis (scverse)

Training and Customization

BioNeMo Recipes:

Distributed parallelism for all foundation model training

Low precision, long context, multi-modal fusion

Deployment

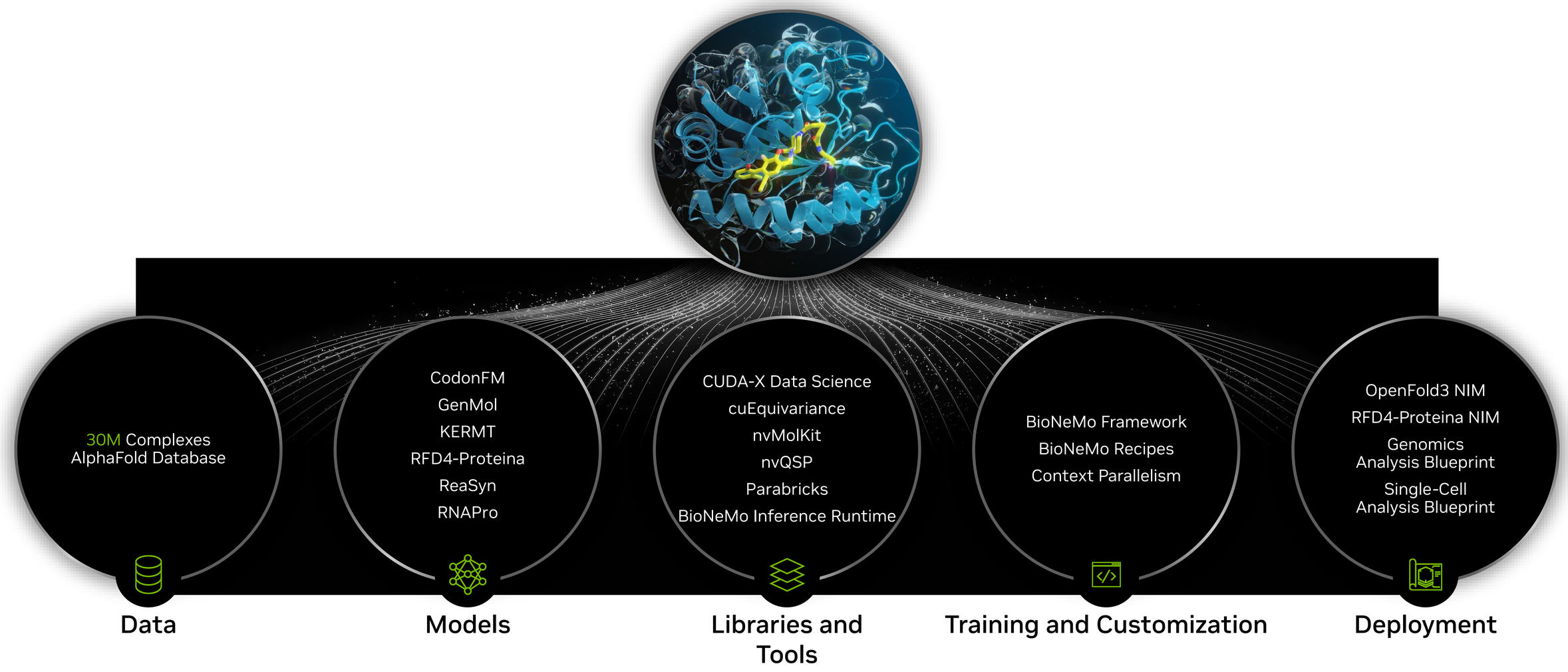
NIMs: Microservice for Enterprise Deployment

BioNeMo Inference Runtime: Inference compiler for bio AI

Blueprints: Reference workflows

NVIDIA BioNeMo

Open models, data and tools for drug discovery dry lab



BioNeMo Agent Toolkit

Get Started Today

- Agent agnostic Library, Open model, datasets and NIM skills
- Agent agnostic workflow skills for lifesciences
- Fully open-source

NVIDIA BioNeMo Agent Toolkit

Turn any agent into a life science expert with NVIDIA BioNeMo skills.

Protein folding, molecular docking, generative chemistry, genomics analysis, protein design, and biomarker discovery — a decade of NVIDIA life sciences libraries, tools, and models, packaged as ready-to-call agent skills.

Each skill gives a coding or scientific agent structured instructions, scripts, and references to select a tool, prepare inputs, run it, inspect outputs, and explain results — across both single tasks and multi-step scientific workflows.

Install

Skills install with the `skills` CLI:

```
# interactive — pick a skill + install destination
npx skills add NVIDIA-BioNeMo/bionemo-agent-toolkit

# one skill, no prompts
npx skills add NVIDIA-BioNeMo/bionemo-agent-toolkit --skill boltz2-nim --yes

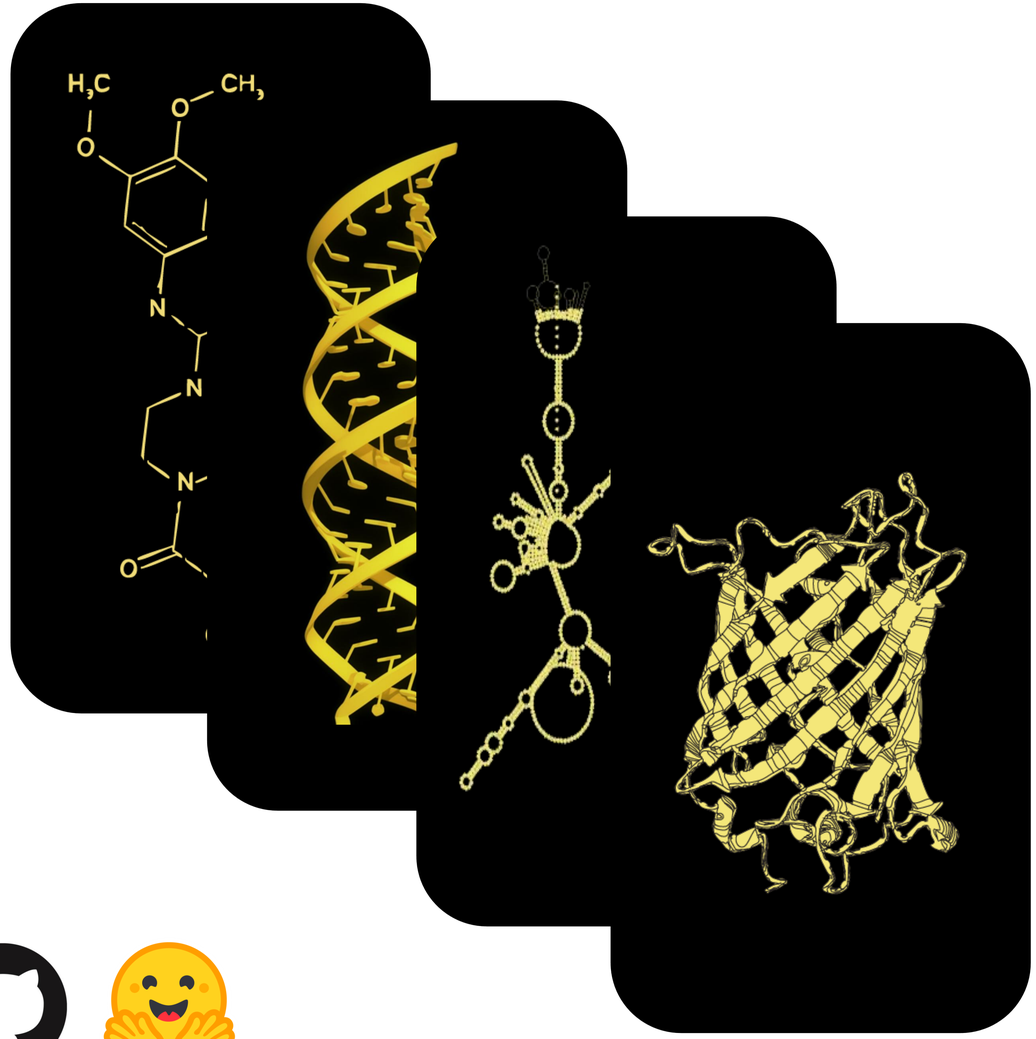
# target a specific agent (repeatable)
npx skills add NVIDIA-BioNeMo/bionemo-agent-toolkit --skill boltz2-nim --agent claude-code
npx skills add NVIDIA-BioNeMo/bionemo-agent-toolkit --skill boltz2-nim --agent codex

# browse the catalog without installing
npx skills add NVIDIA-BioNeMo/bionemo-agent-toolkit --list
```


Source information in NVIDIA Sans, medium gray; 20 pt

NVIDIA

BioNeMo Open Models and Datasets



Understand

codonFM RNAPro

Design

RFD4-Proteina GenMol

Megalodon

Optimize

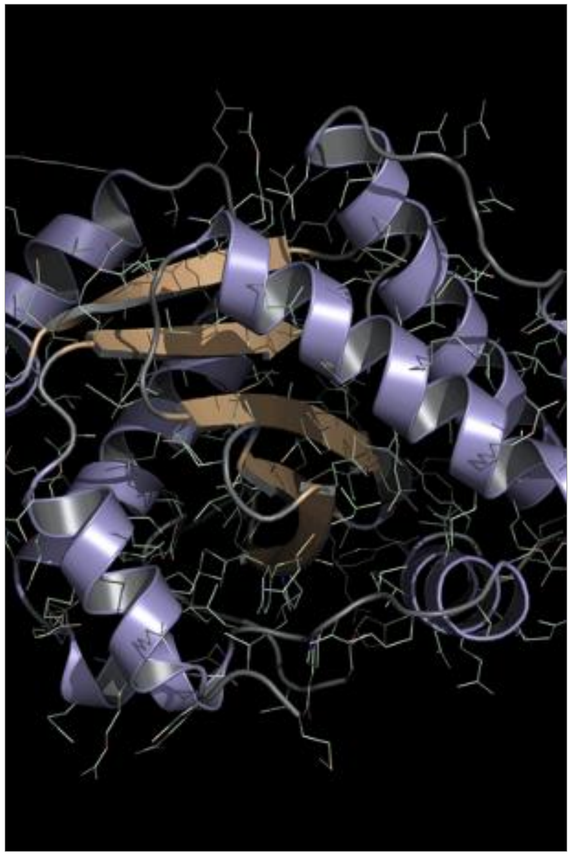
KERMT ReaSyn DualBind

Available on GitHub and Hugging Face

NVIDIA

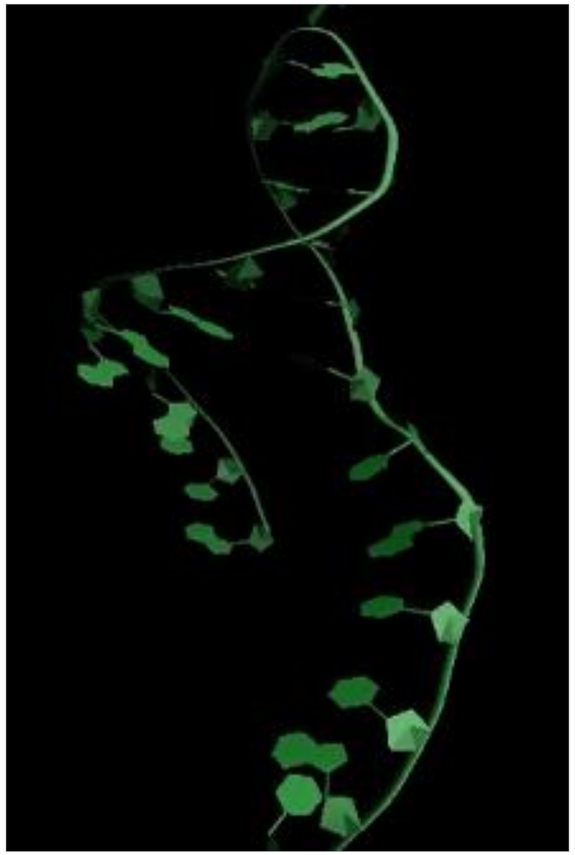
NVIDIA Clara Open Models

Making biology & chemistry programmable



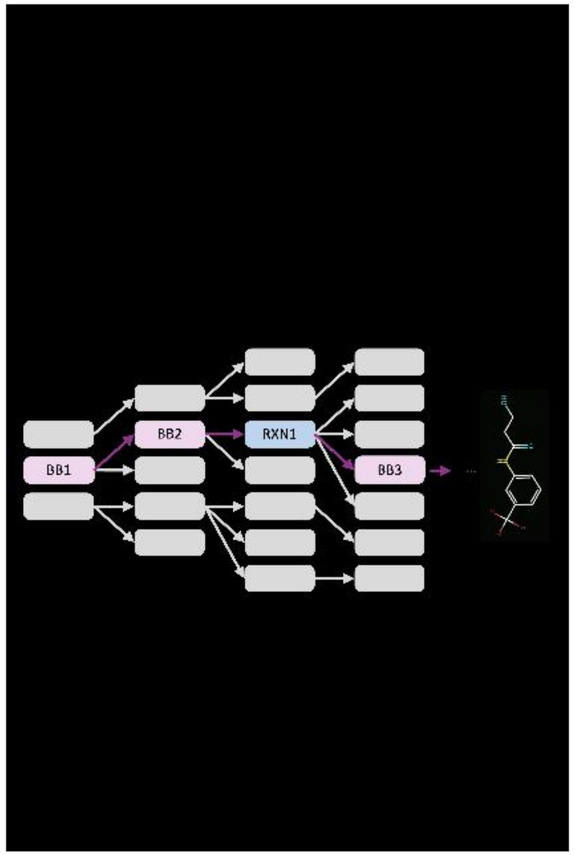
RFD4-Proteina

Creates 3D protein structures for designing medicines



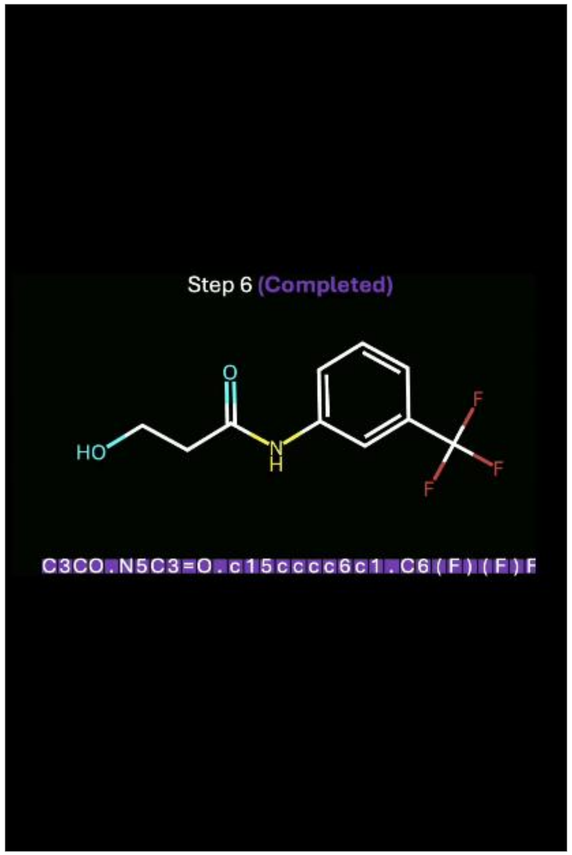
CodonFM | RNAPro

Optimizes RNA sequences for improving therapeutic design; RNA structure



ReaSyn

Designs chemical recipes for making new medicines and materials



KERMT

Pretrained graph neural network model for molecular property prediction.

NVIDIA NIM Optimized Inference Microservices

Rapidly deploy reliable building blocks for accelerated generative AI anywhere



Portable Run cloud-native microservices anywhere, maintaining security and control of data and apps

Easy to Use Move fast with the latest agentic AI building blocks for reasoning, retrieval, images and more, deployed in minutes with standard APIs

Enterprise Supported Gain confidence with stable APIs, quality assurance, continuous updates, security patching, and support

Performance Optimize accuracy, latency and throughput to meet requirements with lowest TCO

Microsoft Azure · aws · Google Cloud · ORACLE · DGX & DGX Cloud · cisco · DELL Technologies · Hewlett Packard Enterprise · Lenovo · SUPERMICRO

NVIDIA

BioNeMo NIM for Lifesciences

Domain-specialized optimized containers for at-scale deployment

At-Scale Deployment for domain specialized use cases:

Protein Structure Prediction, Co-folding with protein, RNA/DNA and small molecules, multiple sequence alignment with GPU mmseqs2

Self-Host or Deploy through Partner platforms :

Download NIM container and deploy anywhere immediately for air-gapped deployment. You can access BioNeMo NIMs through several of our platform or cloud partners.



Boltz 2



OpenFold 2



OpenFold 3



GPU MSA



Evo 2



GenMol



RFD4-Proteina



ProteinMPNN



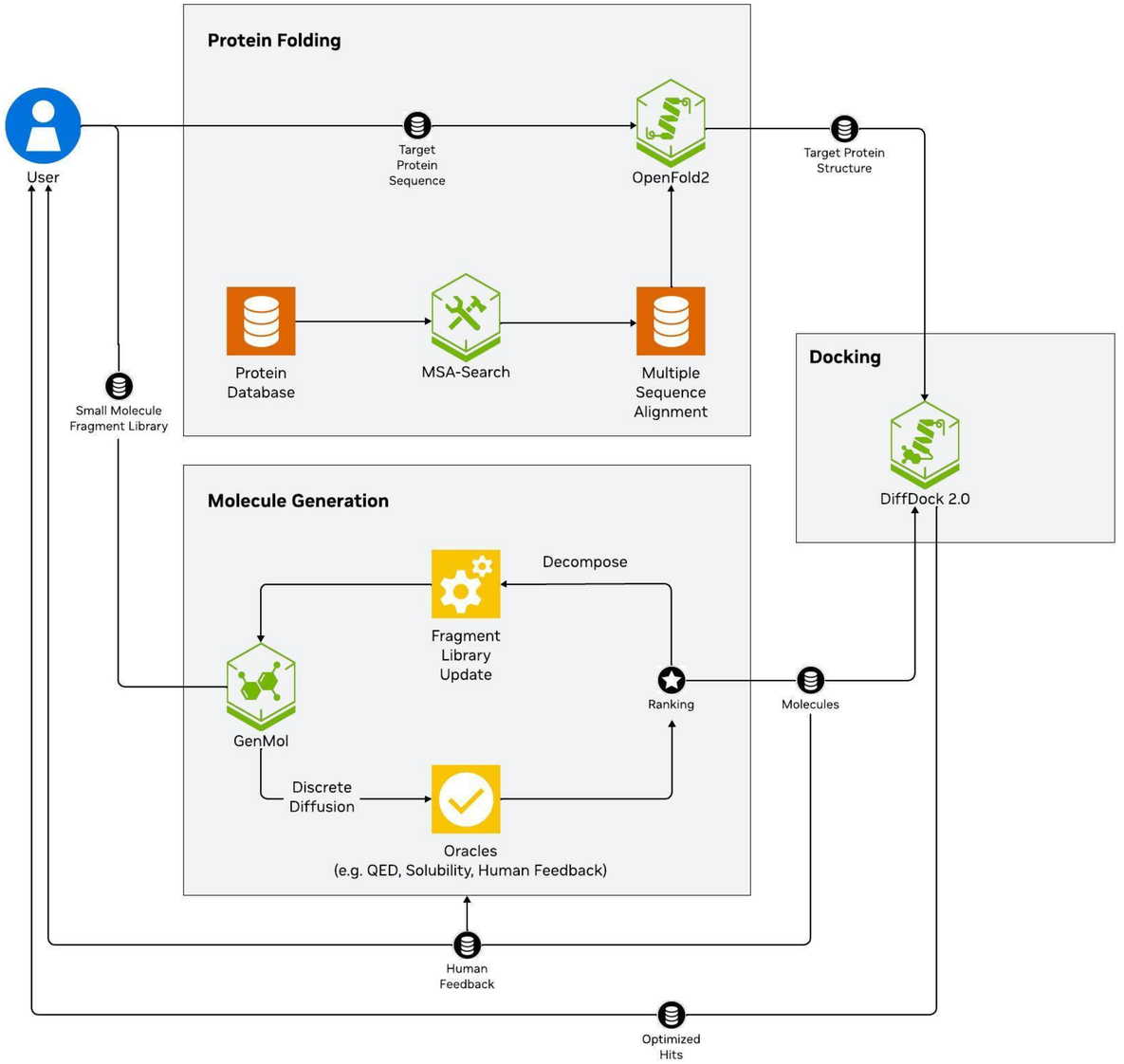
DiffDock

Generative Virtual Screening

Models: MSA-Search, AlphaFold2, GenMol, DiffDock

Benefits

- Use generative AI to more efficiently explore chemical space to optimize molecular designs for multiple features simultaneously
- Accelerated NIMs allow rapid evaluation of large molecule databases to identify better drug candidates faster
- Test fewer molecules to identify virtual hits, reducing the time and cost of drug development



Generative Protein Binder Design

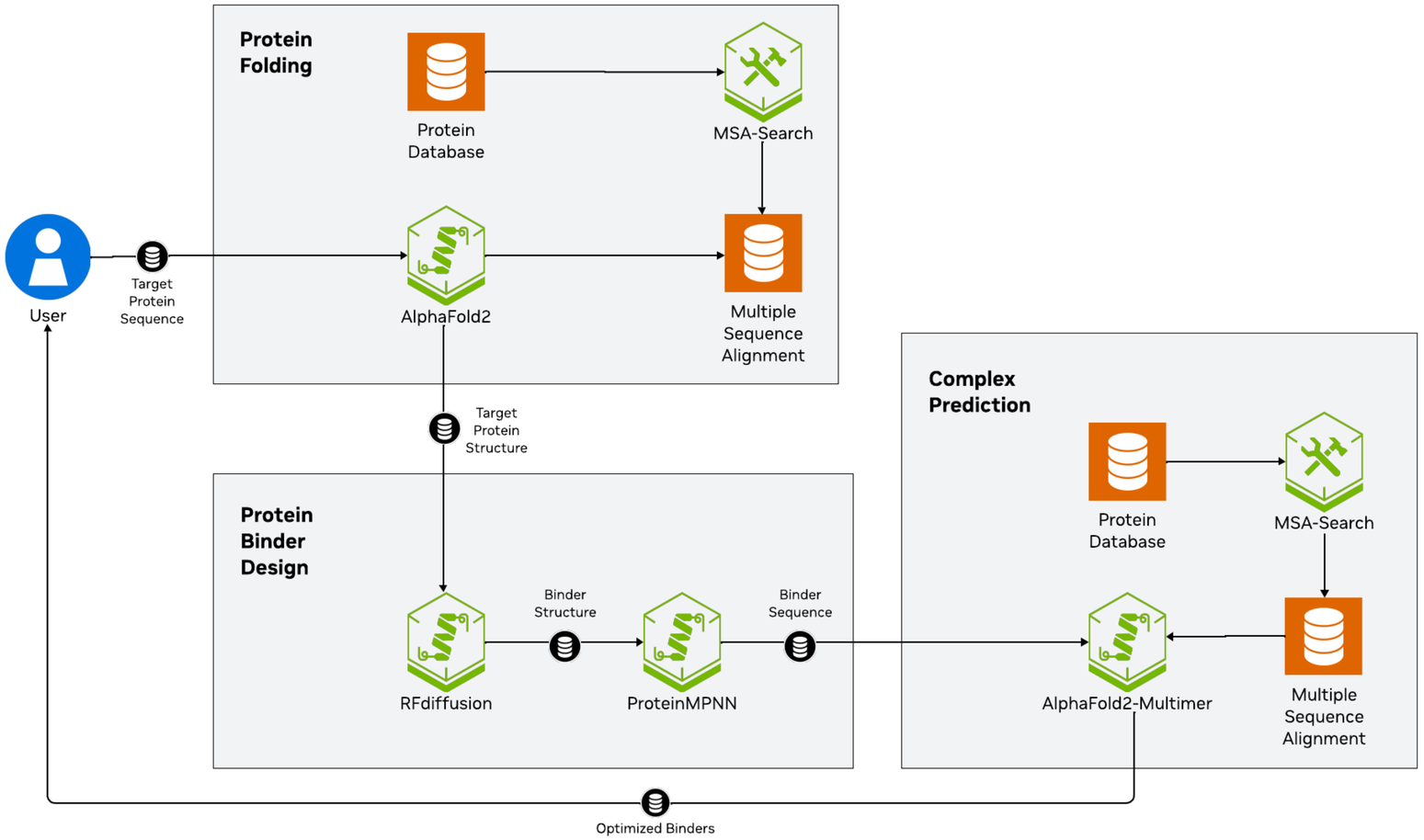
Models: MSA-Search, AlphaFold2, RFdiffusion, ProteinMPNN, Alphafold2-Multimer

Use Cases

- Design novel proteins to bind to a target of interest and/or perform a desired function.

Value Prop:

- Efficiently explore vast protein design space for structures with precise functional features
- SOTA speed and scalability over other approaches
- Identify high-affinity binders from a smaller set of designs



BioNeMo Recipes

Achieve maximum throughput for transformer-based models

BioNeMo Recipes showcases best practices for accelerating biological foundation models using NVIDIA's state of the art CUDA-X libraries and tools.

Core Capabilities

Flexible scaling

Scales from single-GPU prototyping to multi-node training without complex parallelism configurations

Performance optimization

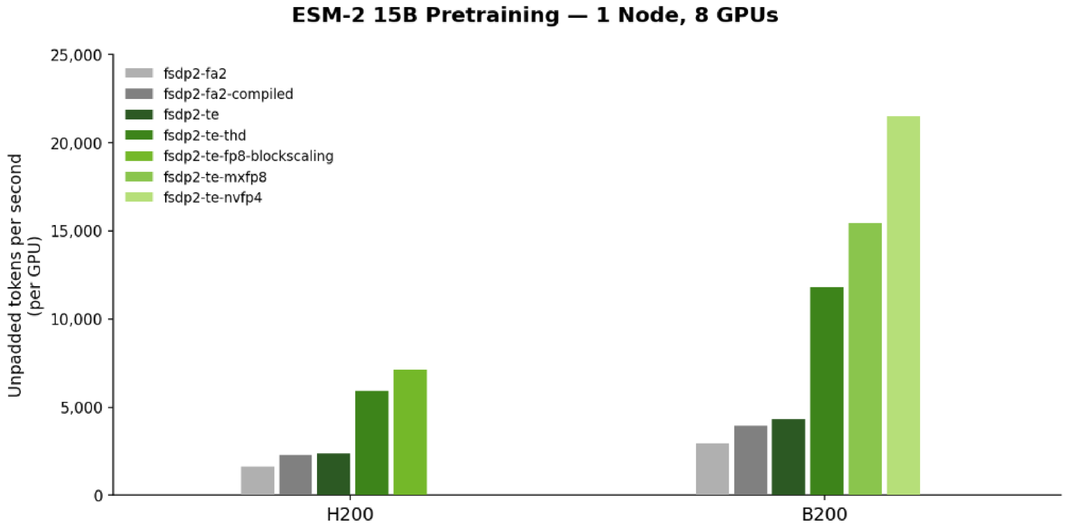
Leverages TransformerEngine and megatron-FSDP for state-of-the-art training efficiency

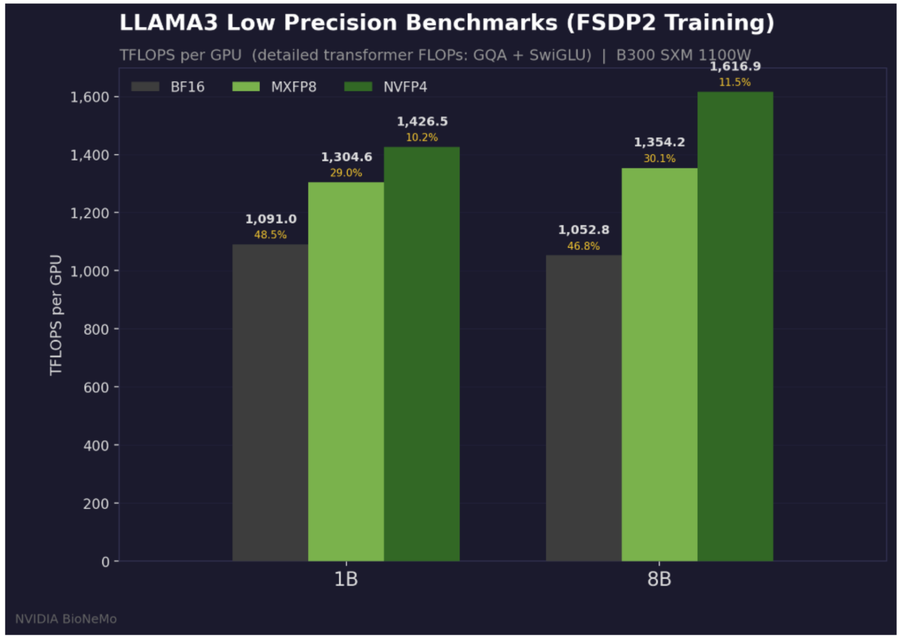
Feature demonstrations

FP8/nvFP4 training, megatron-FSDP, context parallelism, sequence packing

Research-friendly

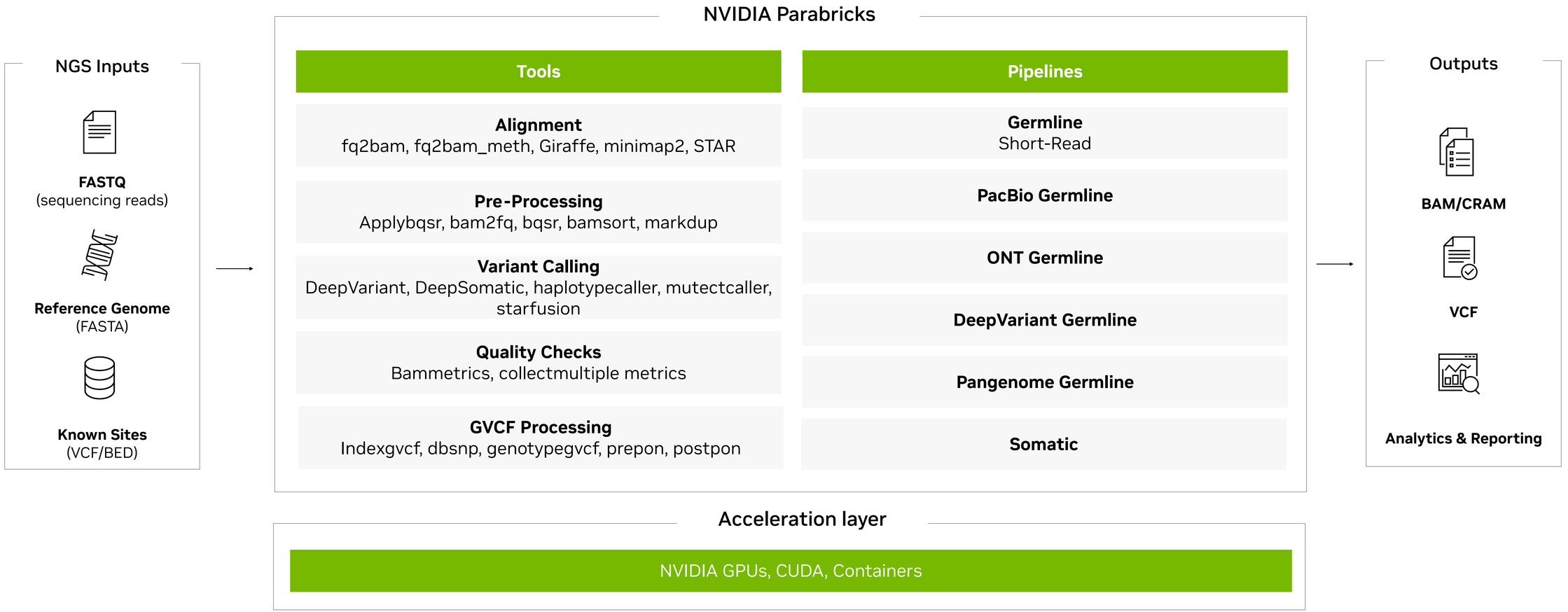
Contains hackable and readable code that researchers can easily adapt for their experiments





NVIDIA Parabricks

GPU-Accelerated Software Suite for Secondary Analysis



BioNeMo Resources

NVIDIA AI Platforms for Healthcare and Life Sciences

NVIDIA BioNeMo — GitHub

NVIDIA BioNeMo Agent Toolkit — GitHub

NVIDIA BioNeMo Inference Runtime

NVIDIA Healthcare AI Models and NIM Microservices

Build an AI Scientist for Life Science Discovery with NVIDIA BioNeMo Agent Toolkit

NVIDIA